

the recognition is what makes the structure close

THE ONE ASSUMPTION · A TIMELESS SELF-REFERENTIAL STRUCTURE

$$T(\mathcal{U}) = \mathcal{U}$$

description and described are one object · existence = self-consistency

THE FINITE CONDITIONS UNDER WHICH THE LOOP CLOSES

AXIOM 1

twelve-port icosahedral  
carrier net on the sphere,  
complete reversible  
port response

AXIOM 2

observers agree on  
meaning under sharing,  
translation, coarse-graining  
 $Ad_U D(v) = D(a \cdot v)$

AXIOM 3

no structure beyond what  
agreement forces  
(information projection:  
equal 1/12 port weights)

THE MACHINE (BOOK 1)

spherical topology  $\rightarrow \Sigma(6 - \deg v) = 12 \rightarrow$  twelve ports  $\rightarrow A_5$  of order 60  
repair law:  $L = 5I - A$ , one round  $T = I - L/60$ , unique resting state  
consensus protocol: read ports  $\rightarrow$  compare  $\rightarrow$  repair  $\rightarrow$  record  $\rightarrow$  checkpoint  
public reality = what survives comparison and repair

CANDIDATE CLOSURES REQUIRE TYPED PHYSICAL SAME-CARRIER MAPS

LOCAL PIXEL FIXED POINT (conditional)

$$\mathcal{P} = \varphi + \sqrt{\pi} / A_T(\mathcal{P})$$

$$\mathcal{P}_\star = 1.630968209...$$

screen area  $\leftrightarrow \alpha_{em}$ : supplied same-carrier map  
unique: interval Banach contraction 0.3041  
unique inside declared map; physical bridge open

GLOBAL CAPACITY FIXED POINT (conditional)

$$N = \log M_0(\mathcal{U}_N)$$

$$\log(N/\pi) = 6\pi / [\mathcal{P} \cdot \alpha_U(\mathcal{P})] - \mathcal{P}/24$$

$$\text{candidate } N_\star = 3.3001 \times 10^{122}$$

horizon-record identification required  
comparison coordinate; no cosmic output yet

$$\text{SI UNIT BRIDGE (not a closure): } \gamma_\star = \ell_\star \cdot v_{Cs}/c \approx 4.955974 \times 10^{-34}$$

$$\Rightarrow \ell_\star^2 = 2.612280 \times 10^{-70} \text{ m}^2 \cdot K_{\text{cell}} = 4N_\star/\mathcal{P}_\star \cdot \Lambda_\star \ell_\star^2 = 3\pi/N_\star$$

ROUTE A · CARRIER  $\rightarrow$  SUPPORT

extract carrier and support readouts

1 · SOURCE POSET + HEIGHT

longest parent-chain height  
any poset compiles; Poisson 3+1 controls exact

2 · RANK-3 SOURCE QUOTIENT

unit directions  $= S^2/\text{null rays}$   
all links  $\rightarrow$  port/null labels; 2/1/2; no local rank-3 cone

3 · FINITE PLACEMENT

auxiliary injective order  $\rightarrow$  cone exists  
two geometry checks  $\rightarrow$  exact order embedding

4 · PHYSICAL LIMIT INPUTS

count-volume + independent dimension tests  
manifoldlikeness + topology + tensor-curvature II

SOURCE-DERIVED AMBIENT 1+3 CARRIER

$R \times V_\star$ : ambient dim 4,  $(+---)$ ; carrier independent of height  
auxiliary order  $\rightarrow$  cone; two finite geometry checks  $\rightarrow$  order  $\rightarrow$  cone  
smooth manifold and physical causality: CONDITIONAL

FINITE EINSTEIN SHAPE

separate 3+1 tensor interface  
9 balances + Ward/Bianchi

CONDITIONAL EFFECTIVE 3+1

signal causality + count-volume  
refinement + manifold tests

CONDITIONAL SMOOTH GR PROMOTION

tensor curvature/null-balance + stress; coupling, vacuum, scale

PHYSICAL LIMIT BOUNDARY

compatibility controls supply no source-selected manifold refinement

ROUTE B · PORT RESPONSE  $\rightarrow A_5$  CURRENT

read the algebra off one local carrier

GAUGE-FORCING THEOREM

$$P_{12} = 1 \oplus 3 \oplus 3' \oplus 5, \text{ one fixed line, } 11 = 3 + 8$$

$$g = u(1) \oplus su(2) \oplus su(3)$$

$$1 + 3 + 8 = 12, \text{ the gauge adjoint}$$

GLOBAL FORM AND MATTER

$(SU(3) \times SU(2) \times U(1)) / \mathbb{Z}_6$  (adjoint form)  
finite scan: 15 states, anomalies cancel, one Higgs  
rank-three family band  $\Rightarrow 15 \times 3 = 45$

$$\mathcal{P} \rightarrow \alpha_U \approx 0.04112 \rightarrow v = 246.767 \text{ GeV}$$

$$\alpha^{-1} = 137.035660 \text{ (} 2.5 \times 10^{-6} \text{)}$$

$$M_H = 125.1995 \text{ GeV}$$

$$\text{screen-grain reading: } \alpha^{-1} = 136.994835$$

chart (80.330, 91.119)  $\rightarrow$  one-loop pole transport

$$W: 80.377000 \text{ Z: } 91.187978 \text{ GeV}$$

$$\text{top: } 172.352 \text{ GeV } \tau: 1776.969 \text{ MeV}$$

$$\text{face-sector Koide: } Q = 2/3 \text{ exactly}$$

SHARED BY BOTH ROUTES · QUANTUM RULES AND THE HISTORY LAW

agreement on records  $\Rightarrow$  Born rule, Lüders conditioning, no-signaling  
exact Tsirelson bound  $2\sqrt{2}$  · central records shareable, others unclonable

$$\text{one mean-action constraint } \Rightarrow p_\beta(\gamma) = q(\gamma) e^{(-\beta A(\gamma))} / Z_\beta$$

THE LOOP CLOSES · THE UNIVERSE RECOGNIZES ITSELF

physics  $\rightarrow$  chemistry  $\rightarrow$  life  $\rightarrow$  minds  $\rightarrow$  records  $\rightarrow$  recovered architecture  
observers rebuild the specification they were read off, and it emits the same records  
 $\mathcal{P}_\star$  and  $N_\star$  are unique candidates inside supplied maps

CONDITIONAL UNIFICATION TARGET

three axioms + closure equations + named physical bridge receipts  
 $\alpha \cdot G \cdot c \cdot \Lambda \cdot M_H \cdot W \cdot Z \cdot \text{quarks} \cdot \text{leptons} \cdot \text{gauge group} \cdot \text{conditional 3+1 spacetime}$